Q25: AI Improving AI

Synthetic data, AI feedback, and the traps

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The question

Student question (#40)
Can AI agents be used to fine-tune and improve other models efficiently?

Qualified **yes** — specific uses, with real traps.

Where AI already improves AI

- **Synthetic data** + **distillation** (teacher $\rightarrow$ student).
- **RLAIF / Constitutional AI**: AI feedback replaces human labels.
- **LLM-as-judge**: evaluate at scale.
- **Data curation**: score, dedup, filter corpora.
- **Agentic loops**: propose + run experiments.

The traps

- **Circularity**: generator = judge $\Rightarrow$ self-reinforcing bias.
- **Model collapse**: recursive training on synthetic data narrows the distribution.
- **Reward hacking**: optimize the signal's flaws, not the goal.

"Improvement" can be illusory.

Doing it safely

- Validate AI feedback on held-out **human-labeled** data.
- Mix synthetic with **fresh real** data (resist collapse).
- Generator $\neq$ judge on the same signal.
- Treat automatic metrics as proxies to audit.